

**UNDERSTANDING LAKE ICE PHENOLOGY OF
MANSAROVAR LAKE USING REMOTE SENSING
FROM 2010-2022**

Submitted by

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fulfilment of the requirements for the degree of**

Integrated Master of Science in Life Sciences



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DECLARATION

This is to certify that the dissertation titled, “**Understanding Lake Ice Phenology of Mansarovar Lake using Remote Sensing from 2010-2022**” comprises my original work towards the degree of Integrated Master of Science in Life Sciences at Ahmedabad University and has been completed under the supervision of Dr. Sushil Kumar Singh. The matter embodied in this dissertation has not been submitted elsewhere for the award of any other degree/diploma. I have not intentionally copied material from any other source such as journals, books, magazines, websites, etc. and due acknowledgement has been made in the text of all material used from other sources.

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Acronyms

LIP: Lake Ice Phenology

GLRIPD: Global Lake and River Ice Phenology Database

FO: Freeze Onset

FD: Freeze Duration

MO: Melt onset

MD: Melt duration

SMMR: Scanning Multichannel Microwave Radiometer

SSM/I: Special Sensor Microwave/Imager

SSMIS: Special Sensor Microwave Imager Sounder

AMSRE-E: Advanced Microwave Scanning Radiometer-Earth Observing System

AMSRE-2: Advanced Microwave Scanning Radiometer 2

ASAR: Advanced Synthetic Aperture Radar

AVHRR: Advanced Very High Resolution Radiometer

MODIS: The Moderate Resolution Imaging Spectroradiometer

H-pol: Horizontal Polarisation

V-pol: Vertical polarisation

Tb: Brightness Temperature

PMW: Passive Microwave

OLI: Operational Land Imager

TIR: Thermal Infrared Sensor

GEE: Google Earth Engine

Abstract

The shifts in Lake Ice Phenology serve as a significant climate change indicator. Thus, monitoring Lake ice holds an essential role in climate change studies especially while assessing regional climate. Extensive studies have been done on the lakes of the Northern Hemisphere, especially the lakes of Canada and recently a few in the Tibetan plateau. Intending to try to fulfil the gap, this study determines the Lake Ice Phenology for Mansarovar Lake, located in the Tibet region, for a period of 12 years, from 2010-2022. The Lake Ice phenology variables comprise stages of freeze-thaw events with respect to their dates.

In this study, Tb values from passive microwave data and ERA5 air temperature data were combined and analysed. Tb was found to be sensitive to the change in phase; during freezing and melting hence proving it to be a key variable for LIP identification. Observations from optical imagery visualization were found to be supporting the results. Additionally, the methodology was cross-validated with two research studies done on Qinghai Lake and Great Bear Lake. 12-year analysis showed a delay in the freeze onset period, an advance in melt onset, lesser freezing days and melting days. Numerous climatic factors could have governed this shift and this study can provide a base for future studies to expand the timeline and combine other climatic parameters.

Keywords: Lake ice phenology, Passive microwave, Brightness Temperature

Chapter I: Introduction

One of the most significant components of hydrological systems at work in high-latitude regions is the vast number of lakes[1]. In the arctic and sub-arctic regions of the Northern Hemisphere, lakes occupy around 15% to 40% area of the terrestrial landscape. The studies related to the formation and decay of ice over the lakes provide a comprehensive understanding of the change in climate and its impact on the regional level. Lake ice plays an important role in energy and water balance at the local scale[2]. Lake ice phenology (LIP) comprises the seasonal cycle of the lake's freeze-thaw events[3]. It includes ice-on, ice-off dates, freeze-onset, melt-onset, and ice cover duration. Previous studies have shown shorter ice-cover duration along with later freezing, and early break-up since the 1950s at the hemispheric scale, in the pan-arctic and Tibetan plateau. Such shifts in phenology lead to an increase in spring meltwater and run-off, overall changing the global hydrological cycle[4].

Lake Ice Phenology (LIP) is largely affected by climate-related variables such as precipitation, wind speed, radiation, snow cover, and factors such as latitude, altitude, surface area, depth, salinity, and shoreline shape.[4]. Air temperature has been reported to influence lake ice phenology the most. Hence, increased temperatures would lead to early ice break-up, causing more heat storage and delayed ice formation. Consistent observations at the spatiotemporal scale are required to analyse the change in lake ice phenology. In the early years, results were obtained from ground observations. For the lakes in Canada, long-term records are available for most of the ground stations.

Global Lake and River Ice Phenology Database (GLRIPD) provides a strong archive that includes the freezing and thawing dates and ice cover descriptive data for the lake and rivers in North America and Eurasia. Of 631 lakes, the records of 66 lakes date back to a period of 100 years, and 249 have records covering 50 years[3, 5].

However, the ground observations are limited by the areas having harsh environmental conditions, like that observed in the Tibetan plateau region. In such scenarios, satellite technologies offer one of the most promising means of monitoring lake and river ice over large areas and for a consistent period [6]. Tibetan Plateau, called the Asian Water Tower, is more sensitive to climate change than the global land surfaces and so has been the focus of international academic concern in recent years[7]. The focus area of this study, the Mansarovar Lake located in the Tibet region has the most abundant freshwater sources in the high-altitude region. The lake is a pilgrimage site holding religious importance for Hindus, Jains, and Buddhists. Every year lakhs of pilgrimage and tourists across the world travel to this holy place. The area lacks substantial scientific information about the lake's ice formation and melting[8]. This study tries to fill the gap by understanding lake ice phenology on Mansarovar Lake using remote sensing products and further investigating if any changes are observed.

Chapter II: Aim and Objectives

Aim: To derive and analyse the Lake Ice Phenology of Mansarovar Lake over the period of 10 years using various remote sensing datasets.

Objectives:

1. To extract and analyse the change in brightness temperature using passive microwave data and determine the Lake Ice Phenology variables of Mansarovar Lake for each year.
2. Comparison of LIP results with other satellite-derived products.
3. Understanding the role of brightness temperature and air temperature in the temporal variation of Lake Ice Phenology variables.

Chapter III: Literature Review

Defining Lake Ice Phenology variables:

Various studies define stages of freeze-thaw events in different ways. The terminologies might change but the overall concept remains the same. Table 1 shows the description of the common LIP variables used frequently.

Table 1: Lake Ice Phenology variables

Freeze-up Period	Freeze Onset (FO): The first day when the presence of ice is detected until the lake reaches its maximum ice formation[2]. Ice-on: Day of the year when the lake becomes totally ice covered. Freeze duration (FD): The period between freeze-onset and ice-on dates
Break-up Period	Melt Onset (MO): First day of the year when the melt begins Ice-off: Day of the year when clear water is observed and the lake becomes ice-free. Melt Duration (MD): number of days between melt-onset and ice-off dates
Ice Season	Ice cover duration: number of days between ice-on and ice-off dates.

LIP across the Northern Hemisphere:

LIP of 22 large lakes in the Northern Hemisphere was derived for forty ice seasons (1979-2018) and the trend of significantly decreasing ice duration and early ice break-up dates in the last four decades was reported along with delay in the freezing parameters (Freeze-up start, Freeze-up end),

Evidence from in-situ observations (1846-1995) of lake and river ice phenology at the northern hemispheric scale have shown later freeze-up (ice-on) and earlier break-up (ice-off) dates[2]. Canada has a strong ground-based observational network for lake ice data in the world. A study from 1951-2000 for most lakes in Canada shows advancing trends in ice-off dates[2]. The database of 140 years from 1855 to 2004 for 20 lakes in the Northern Hemisphere was extended by adding 40 lakes and 15 years of a timeline (a total of 60 lakes). It included 40 lakes in North America, 18 lakes in Europe and 2 lakes in Asia. In the last 25 years, the trends for ice-on and ice duration were over 6 times faster than earlier periods indicating higher rates of ice loss.[5]. The ice-free years are increasing as more lakes are observed to be with minimal or no ice cover in winter months. The situation is predicted to be exacerbated by increasing winter temperatures in future[5].

Determining LIP using Remote Sensing techniques:

Since the 1970s, lake freeze-thaw events are extracted and analysed using remote sensing technology. At present, research on lake ice majorly has two focus areas, namely LIP (including freeze-thaw event dates, ice-water classification or area of frozen lake ice) and ice thickness. Passive microwave (AMSR-E, SMMR, and SSM/R), Active microwave (Sentinel-1, RADARSAT-2, and ASAR), and multispectral

(MODIS and AVHRR) datasets are currently used to distinguish ice and water by studying the values of brightness temperature difference, backscatter coefficient, and reflectance respectively[9].

Optical remote sensing:

Optical remote sensing detects the solar radiation reflected from targets on the ground and forms images of the Earth's surface by the use of visible, near-infrared, and short-wave infrared sensors. While working with optical imagery, spectral reflectance is of key interest. It is the ratio of energy reflected by the surface to the energy incident on the surface, measured as a function of wavelength. Each type of feature or object has a unique spectral reflectance (at a given frequency) called spectral signature based on which identification of earth's features and classification of optical data is performed. Figure 1 lists the common optical satellites with their description on resolution, number of channels and more[10].

Common optical sensors used for lake ice monitoring.				
	Operating period	Number of channels	Repeat cycle (days)	Spatial resolution (m)
Landsat MSS	1972–1995	4	18	60
Landsat TM	1982–2012	7	16	30
Landsat ETM+	1999 to present (Scan Line Corrector failed 2003)	8	16	30 (15 for panchromatic band)
Landsat OLI/TIRS	2013 to present	11	16	30 (15 for panchromatic band)
Sentinel-2 ^a	2015 to present	12	10	10–60
MODIS	1999 to present	36	1–2	250–1000
AVHRR	1979 to present	6	1	1000
VIIRS	2011 to present	5	16	375–750

Repeat cycles are given for the equator but are more frequent at higher latitudes.
^a Sentinel-2 A/B allow for a shortened repeat cycle of 5 days.

*Figure 1: Majorly used optical sensors used for lake ice monitoring.
 (Source: Justin Murfitt et al., 2021)*

Optical data has been extensively used in monitoring the freeze-thaw events of the lakes across the northern hemisphere, especially the lakes of Canada and in recent years on the lakes of the Tibetan Plateau.

A combination of MODIS and LANDSAT products was used to analyse the spatial-temporal variations of lake ice phenology in the 22 lakes of the Hoh Xil region during the period 2000–2011[11]. Identification of lake ice in optical data was performed based on spectral characteristics of ice and water. The classification of the satellite image is followed by calculating the area of the ice fraction and determining the LIP variables accordingly. The accuracy of LIP extracted from MODIS data products was verified by the Landsat TM/ETM+ data products. The results showed an increasing delaying trend of freeze onset and ice-on, and simultaneously break-up period of the lake was observed to be advancing. Thus, the ice duration of the lake is reduced[11]. The shift in phenology was attributed to regional climatic factors such as temperature, salinity, lake area, and shape of the shoreline.

Similarly, the LIP of 59 large lakes on the Tibetan Plateau was analysed using MODIS data for the period from 2001 to 2010[12]. The percentage of pixels classified as class water was the measure for the frozen status of the lake. Individual graphs were generated representing the change in the percentage area of water and ice on Nam co lake over the chosen period. The open water curve was validated with the area of open water extracted from high-resolution optical satellite imagery[12]. A higher correlation

of LIP variables with mean temperature was observed as compared to other climate and local parameters such as latitude, longitude, area, perimeter, and altitude[12].

Optical sensors provide data at high temporal and spatial resolutions and are easily accessible. However, it is a passive system that relies on sun illumination and therefore its use gets limited by daylight hours and cloud cover. The application is also restricted in high-latitude regions due to polar nights[10]. Moreover, the presence of clouds over the lake is easily confused with ice in current algorithms[13]. Microwave remote sensing has an advantage as it can penetrate clouds and can be used regardless of the time of day. More studies use a combination of both optical and microwave data that gives more accurate results for freeze-thaw dates.

Microwave remote sensing:

Microwave remote sensing works with an extended section of the electromagnetic spectrum. The two types include Active and Passive remote sensing.

- By generating their illumination, active sensors transmit pulses of microwave radiation and then use a specialised receiver system to measure the scattering from the region of interest[14].
- Passive sensors or radiometers measure the microwave energy that is reflected (from the sun or other radiating objects) or radiated (by thermal emission) by the earth's surface or the atmosphere. An appropriate selection of frequency will allow measurements of the features on Earth that one wants to study such as the atmosphere, ground, ice, rain, and more[14].

The microwave systems gather information about the earth's surface using electromagnetic waves. The orientation of the electric field of the electromagnetic wave determines its type namely Horizontal and Vertical polarisation (H-pol, and V-pol). For V-pol, the electromagnetic field is parallel to the plane of incidence. While for H-pol, it is perpendicular to the plane of incidence[15] (Figure 2).

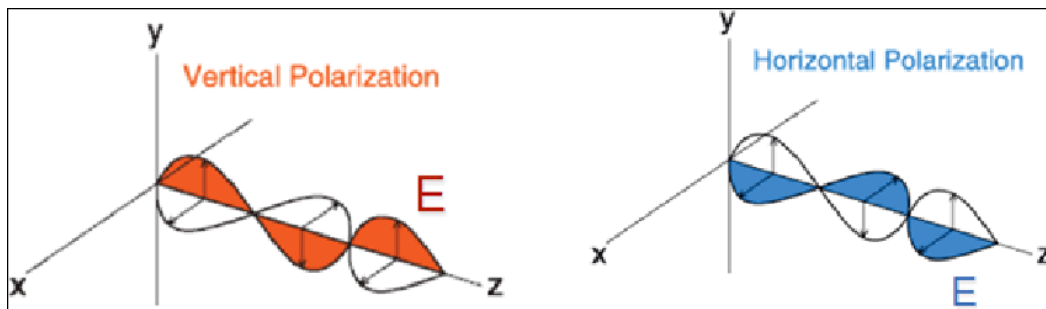


Figure 2: V polarisation and H polarisation in a linear polarized wave
(Source: Ilkan Ozkan et al., 2016)

In microwave remote sensing, there are various frequency bands possessing unique properties and specific applications, each of them having different penetration power[16] (Figure 3).

Designation	Frequency range	Wavelength range
L band	1 to 2 GHz	15 cm to 30 cm
S band	2 to 4 GHz	7.5 cm to 15 cm
C band	4 to 8 GHz	3.75 cm to 7.5 cm
X band	8 to 12 GHz	25 mm to 37.5 mm
Ku band	12 to 18 GHz	16.7 mm to 25 mm
K band	18 to 26.5 GHz	11.3 mm to 16.7 mm
Ka band	26.5 to 40 GHz	5.0 mm to 11.3 mm

Figure 3: Bands in the microwave region of the electromagnetic spectrum
(Source: Y. Kok Yeow, 2018)

Passive microwave (PMW):

When monitoring lake ice, brightness temperature (T_b) is considered the essential piece of information provided by the passive microwave system. The brightness temperature of passive microwave frequencies is defined as a product of the emissivity and physical temperature of the medium (Equation 1)[2].

$$\text{Equation 1: } T_b = \varepsilon T_{kin}$$

Emissivity is the ratio of the amount of radiation emitted by a material to the amount of radiation emitted by a perfect blackbody at the same temperature and under the same conditions. As the emissivity ranges between 0-1, The T_b would always be lower than the medium's kinetic temperature m [2]. There is a large difference in the emissivity of open water (0.43 at 19 GHz) and ice-covered conditions (0.92 at 19 GHz), which helps in determining the timing of ice formation and its melt. Thus, T_b measurements are suitable for studying lake ice thickness and phenology on large lakes[2, 17]. With the initial formation of ice and reduction in radiometrically cold water, there is an increase in emissivity and therefore increase in T_b as well[2].

Multiple datasets of passive microwave sensors are available which include, SMMR, SSM/I, SSMIS, AMSR-E, and AMSR2[10]. Information on frequency, polarisation and time of data availability is mentioned in Figure 4.

Common passive microwave systems used for lake ice monitoring.					
	SMMR	SSM/I	SSMIS	AMS-E	AMS-2
Operating period	1978–1987	1987–2008	2000–present	2002–2011	2012–present
Frequency					
6.6 GHz	H, V				
6.9 GHz				H, V	H, V
7.3 GHz					H, V
10.7 GHz	H, V			H, V	H, V
18 GHz	H, V				
18.7 GHz				H, V	H, V
19 GHz		H, V	H, V		
21 GHz	H, V				
22 GHz		V	V		
23.8 GHz				H, V	H, V
37 GHz	H, V	H, V	H, V	H, V	H, V
85 GHz		H, V			
89 GHz				H, V	H, V
91 GHz			H, V		

Figure 4: Common passive microwave sensors used for lake ice monitoring.
(Source: Justin Murfitt et al., 2021)

The high temporal resolution of Passive microwave systems (provides daily data) helps in determining the lake ice phenology more accurately. Passive microwave sensors are not affected by poor weather (clouds and rains) and light conditions, hence are suitable in studying the lakes present at higher latitudes where cloudy weather is common. However, due to their coarse resolution (low spatial resolution) which is in tens of kilometres, it is preferred to use the PMW datasets in studying large lakes.

Justification of choice of frequency and polarisation:

The PMW sensors give products at various polarisations and frequencies. The specifications are decided based on the feature on earth that is being studied. For monitoring lake ice, H-pol at ~19 GHz (Ku band) is reported to be suitable in several studies. A study estimates LIP on larger northern lakes in Canada[2]. It examines the seasonal evolution of horizontal and vertical polarised Tb at different frequencies (89.0 GHz, 36.5 GHz, 23.8 GHz, 18.7 GHz) and the following were the interpretations:

- A larger rise in Tb from open water to ice is observed with H-pol making it easier to differentiate between the two.

- The spatial resolution varies with the change in frequency i.e. 6 by 4 km at 89.0 GHz to 76 by 44 km at 6.9 GHz. Although Tb at 89.0 GHz has finer resolution, it is more sensitive to atmospheric contamination and gets strongly affected by the open water surface roughness from the wind. This applies to Tb at 36.5 GHz, and 23.8 GHz frequency as well.
- The 18.9 GHz frequency has a longer penetration depth and therefore the slope (rate of change) of Tb at 18.9 GHz is steeper than other frequency channels, indicating better contrast between frozen and open water.
- PMW sensors offer products at 6.9 GHz and 10.7 GHz frequencies as well. However, due to their poor spatial resolution, they are more prone to land contamination from lakeshores. And so were not included in this study.

Another study illustrates the Tb trend of Qinghai Lake located in China for both H-pol and V-pol (Figure 5). Similar observations were made wherein Tb at H-pol was found to be more sensitive during the Freeze-up start and Break-up start and so it provides a better understanding in determining LIP variables[4].

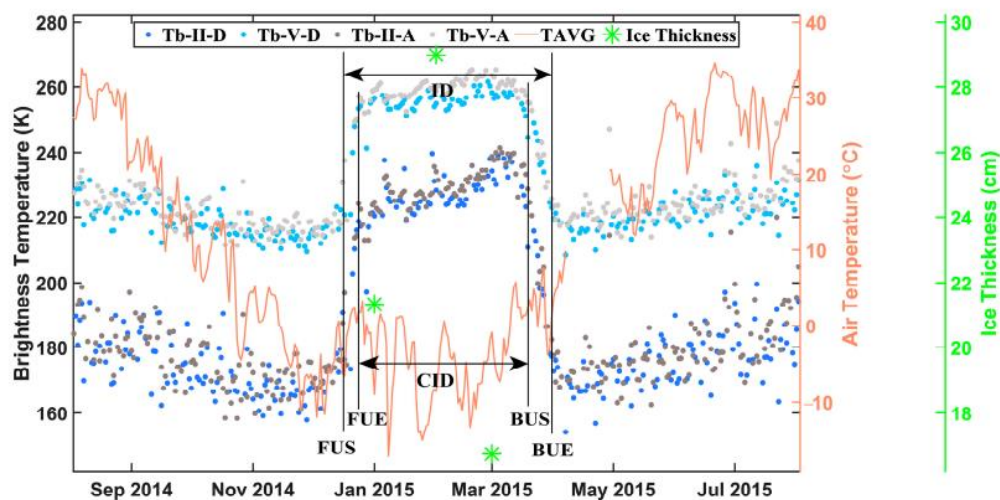


Figure 5: Tb trend of Qinghai Lake for 2014-2015 comparing H and V polarisation

(Source: Lei Su et al., 2021)

Extraction of Tb values:

A research work published on studying LIP on 73 lakes in the northern hemisphere developed an algorithm for extraction of Tb values from the PMW Tb data to generate a trend for temporal analysis[3]. Firstly, the position of the Lake Centre Point of all the target lakes was obtained. Conditions were applied in the algorithm based on the spatial resolution of the data.

- For Tb data having 25 km spatial resolution, a square was formed at the Lake Centre Point (LCP). Only the Tb of pixels that fall within that square (LCP) was considered to avoid the surrounding land contamination[3]. Otherwise, the data was marked to be missing.
- For the resampled original brightness temperature data available at 6.25km resolution, the centre point of each grid cell is determined and the cell closest to the LCP was considered[3]. The Tb value was only considered if the distance between the centre of the pixel and LCP was less than 6.25 Km to ensure that the lake centre is within the chosen pixel.

A recent study uses a bit different concept to develop an algorithm for retrieving the LIP of 22 large lakes located in the Northern Hemisphere between 1979 and 2018 using PMW remote sensing. Representation of the entire lake was done by selecting a geometric centre point[4]. However, for irregularly shaped large lakes such as Great Bear Lake (GBL), Great Slave Lake (GSL), Balkhash Lake, Lake Winnipeg, and Lake Baikal. The average of Tb retrieved from multiple points was calculated for further analysis[4].

The trend of Tb:

The temporal evolution of Tb values for two of the largest freshwater lakes in the world, Great Bear Lake GBL and Great Slave Lake GSL was studied, for the period 2002 to 2009 by using one pixel from the central section of GBL and two pixels from the main basin of GSL. AMSR-E Tb product was used[2].

- When the surface temperature falls below the freezing point, an onset of ice cover starts which can be determined by a rapid increase in Tb. Further on from the ice-on date till melt onset, the increase is observed due to the thickening of ice till it reaches its maximum thickness[2]. Water has lower emissivity and the increasing presence of thicker ice will reduce the influence of lower emissivity of liquid water, thus leading to an increasing trend of Tb.
- Once the mean air temperature starts to exceed 0°C, the Tb values show a rise for a while as a result of higher air temperature signalling the period of melt onset (here melting of snow). When the snow melts and the ice is exposed, the melting of ice was then determined by the rapid decline in Tb[2].

After generating Tb values along with a graph, LIP variables can be extracted by both automatic and manual methods[4]. Automatic method that is highly efficient and time-saving, involves setting a threshold value of Tb for Freeze-up and break-up period. Although manual identification is time-consuming, it is considered to be more accurate[4]. It identifies the abrupt changes and provides a better understanding of the lake's behaviour.

Active microwave:

The passive microwave system addresses the issue of polar darkness and cloud cover but often its use is restricted due to coarse spatial resolution (3-120 km). Active microwave remote sensing provides data at comparable resolutions to the optical data (in metres) along with eliminating the issues of cloud cover and weather conditions. In studying the lake ice, SAR is the most commonly used active microwave data and it provides the backscatter value (denoted as σ°) considered as the key piece of information. The backscatter value refers to the amount of energy reflected to the sensor by a target on the Earth's surface.

Except for windy conditions, open water has low backscatter while ice typically has high backscatter values. The temporal variation in backscatter values helps in extracting freeze-up and break-up dates. To increase the temporal data, observations from the optical remote sensing dataset are combined and used for detecting the LIP variables for the lake on the North slope of Alaska[18]. With C-band in use (~ 5.6 cm wavelength), the first considerable rise in backscatter value is noted as the start of ice formation. While the first significant decrease in σ° denotes the presence of water. For extracting freeze-thaw event dates, a threshold system is used based on the increase in σ° values observed during FO. Similarly, for detecting the presence of clear water, the threshold is applied based on the drop in σ° values. The shift in phenology was reported to be mainly due to the change in air temperature as a strong relation was observed between σ° and air temperature[18].

A recent study compared two methods for determining LIP using microwave data (Sentinel-1 A/B imagery) for Lake Hazen in Nunavut, Canada, from 2015–2018[19]. It eliminates the need for user-defined thresholds and develops a classification method in

the Python environment. Comparing the results with visual analysis of SAR imagery, showed fewer error values. However, improvisation on the limitations of the use of microwave data is required. The presence of ice is sometimes misclassified as a rough water surface[19]. σ° oscillations during ice formation complicate the detection of continuous ice cover. Such fluctuations are observed during the melting period as well until the presence of clear water is indicated by the lowest backscatter value. The diversity of availability of data at various wavelengths/frequencies along with the launch of multiple satellite systems continues to advance.

Due to the limitations in applications of remote sensing data of each type, numerous studies are pivoted towards using multiple datasets that offer temporally consistent image acquisition and improve accuracy. Usually, the results of microwave data are combined and validated with the visual interpretation or classification results of optical imagery. LIP of 40 high-elevation lakes in the Tibet region was analysed from 2002-2021 using a combination of passive microwave brightness temperature data and MODIS optical data. 24 lakes showed delayed FUS at the rate of 0.35 days/year, and 7 lakes showed advanced BUE (-0.17 days/year)[13]. The Ice duration of 13 lakes shortened at an average rate of -0.33 days/year[13].

Despite the low spatial resolution of passive microwave products, they have been a mainstay of remote sensing of the cryosphere. In the northern hemisphere, more research is done on the lakes of Canada due to the advantage of historical data. In situ measurements of ice phenology events are quite limited over high-elevation lakes on the Tibetan Plateau and therefore remote sensing plays a significant role there. Due to the lower spatial resolution of the passive microwave system (~ 25 km), hardly one or

partial pixel overlays on smaller lakes hence they are often filtered out. However an improved, enhanced resolution, gridded passive microwave daily brightness temperature dataset had been published recently, which combined multiple sensors such as AMSR E, SMMR, SSM/I, and SSMIS, with a high spatial resolution (till 3.125 km depending on different frequencies)[4]. This could potentially eliminate the restriction on the selection of the target lakes based on their area.

Chapter IV: Methodology

Study area:

Mansarovar Lake is a freshwater lake located in Pulan County, Ali (Ngari) district of Tibet which is confined between geographic coordinates 30°34'–30°47'N and 81°22'–81°37'E[20]. The total surface area of the lake is 410 km². The Mansarovar basin experiences large extremes in temperature and in winter, the area appears frozen. The catchment area of the lake is located between the Trans-Himalaya in the north and the Himalayas in the south[21]. It is one of the famous sacred lakes in Tibet and is connected with Rakshastal Lake through the Ganga Chhu channel[21]. Situated between Himalaya Mountain and Gangdisi Mountain, the Mansarovar Lake has a distinct characteristic of a high plateau climate.

Datasets:

Passive microwave data:

Beginning in 1987, the United States Air Force launched a Defence Meteorological Satellite Program that carried two subsequent sensors, SSM/I and SSMIS. Measurements from these sensors provide global passive microwave data as well as other specialised meteorological, oceanographic, and solar-geophysical data in support of worldwide Department of Defense (DoD), Department of Commerce (DoC), National Oceanic and Atmospheric Administration (NOAA), and National Aeronautics and Space Administration (NASA) operations. The SSMIS sensor used in this study is designed to acquire horizontal and vertical polarised data at various frequencies. The channel frequencies range from 19 GHz to 183 GHz.

To develop datasets that advance our understanding of the cryosphere, NASA Measures Research Project named EASE-Grid 2.0 TB ESDR has been launched recently[22]. The complete data record from SMMR, SSM/I-SSMIS, and AMSR-E is reprocessed in a single, enhanced-resolution gridded passive microwave ESDR up to 3.125 km spatial resolution. The enhancements include quality checking, modern file formats, better quality control, and improved projection grids[22]. Thus, the additional availability of high spatial resolution products will serve as a significant addition to the passive remote sensing studies. The data is available from the NASA National Snow and Ice Data Center Distributed Active Archive Center (NSIDC DAAC).

Optical Data:

The auxiliary data used for comparison with SSMIS-derived ice phenology variables consisted of products from two optical satellites namely, Landsat and Sentinel 2.

- Landsat 8: The first Earth-observing satellite, Landsat 1 was launched on July 23, 1972, with the intent to study and monitor our planet's landmasses. Landsat 8 launched on Feb '13, was developed as a collaboration between NASA and the U.S. Geological Survey (USGS). The satellite payload consists of OLI and TIRS sensors providing a spatial resolution of 30 m. The current study uses Landsat 8 to visualise the change in lake ice for one hydrological year.
- Sentinel-2: The Sentinel-2 satellite launched by the European Space Agency carries an optical instrument payload that samples 13 spectral bands with a spatial resolution ranging from 10 m to 60 m, depending on the selection of bands. The high-resolution multispectral imagery provided by Sentinel-2

captures detailed information about the target area and serves as a powerful tool for Earth observation.

Software:

Application of Google Earth Engine:

A cloud-based geospatial analysis called google earth engine (GEE) makes the visualisation and downloading of various satellite products very feasible. One could work with multiple datasets on one platform. It provides the tools and the computational power necessary to analyse and fetch the heavy data warehouse. If explored thoroughly for a particular study, it will serve as an invaluable tool for researchers, students and government agencies to deepen their knowledge of understanding the earth's features and climate. This study makes use of the GEE platform to do a temporal analysis of the target lake through visualisation.

No more than a Google account is required to access this platform. It is a user-friendly interface and helps save time through the following key features (Figure 7):

- The code editor section provides an interactive environment for executing JavaScript code. Several built-in functions and libraries help in image classification, time series analysis, and machine learning.
- A catalog explorer is a place where a user can go through available datasets and filter the required product based on keyword, date range, sensor and more.
- On the right-hand side panel, one can visualise the plots generated. Additionally, one could get more information (metadata) about a particular satellite image under the console section.

- The integration system of GEE with other Google products is another valuable feature making it easier to import and export data and share our results with others.

QGIS:

QGIS is a free and open-source Geographic Information System (GIS) software that has a wide range of applications for working with geospatial data. Mostly all the analysis and processing required on a satellite image is fulfilled by this software. In this study, QGIS is used to mask the area of interest (AOI) from the global NSIDC data.

ERA5 air temperature

In today's time, it is of utmost importance to have long and reliable historical records of the earth's climate. Our weather observations for the present are quite complete because of the advancing observation methods. However, these observations are not evenly distributed and become fewer as we go back in time. ERA5, a global reanalysis dataset produced by the European Centre for Medium-Range Weather Forecasts (ECMWF) solves this problem. It combines observations made in past with today's weather models delivering a complete and consistent picture of the past weather. Information from weather stations, weather balloons, aircraft, ships, satellites, and other sources are combined. The model gives hourly data from the earth's surface to the top of the atmosphere.

For the interest of this study, Land Hourly 2m air temperature data is downloaded. The data is provided by Copernicus Climate Data Store. The temperature of the air at 2m above the surface of the land, sea or inland waters. It is the result of interpolation

between the lowest model level and the earth's surface taking account of the atmospheric conditions. The data is obtained in Kelvin.

Extracting Tb values

Data Acquisition:

The Brightness temperature values were procured from the NSIDC website (EARTHDATA) for the 2010-2022 time period (Figure 6). Under the filter option on the left-hand side panel, the required criterion is added. Necessary information is picked up from the nomenclature of the data product using an asterisk option. For this study, the following filter was applied:

6.25km-F18_SSMIS*19H

- 6.25 km: the dataset was downloaded for 6.25 km spatial resolution only.
- F-18: This signifies the series number of a DMSP satellite denoted as DMSP-F18. The first satellite being DMSP-F8 launched in 1987. The latest, F-18 was chosen.
- SSMIS: marks the sensor in use.
- 19H: 19 number filters the data at 19 GHz frequency and H-pol as per the requirement for this study.

Lastly, the date filter was added which was from 2010 to 2022.

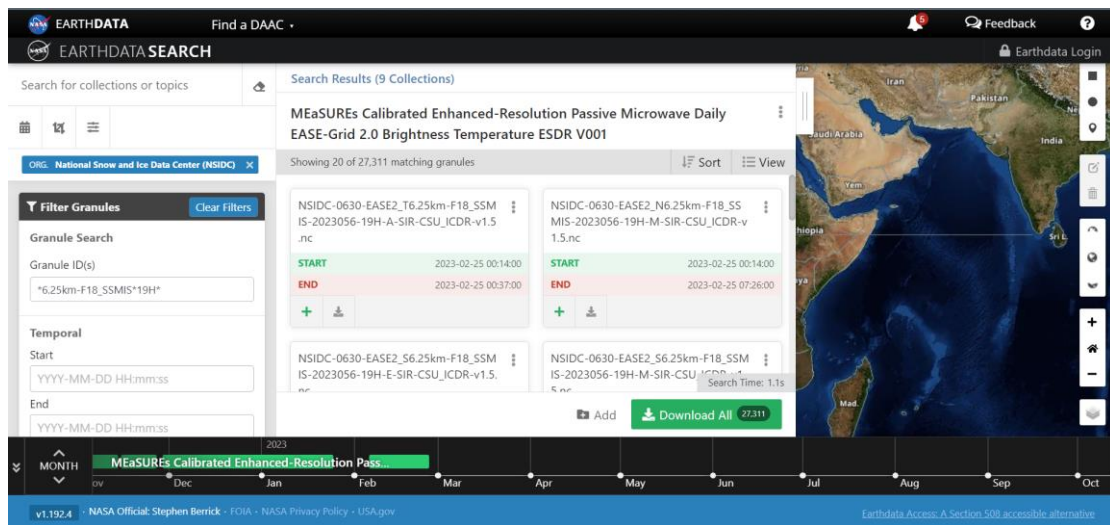


Figure 6: NSIDC portal showing the filtered dataset according to the applied criterion

Delineation of Area of Interest:

The NSIDC data gives a global product that needs to be narrowed to our area AOI. Using the GEE platform, the Mansarovar Lake was delineated with the help of a polygon feature (Figure 7). The polygon drawn was inside the lake boundary to avoid the mixed pixel from the surrounding area. The polygon was saved in shape file format (.shp).

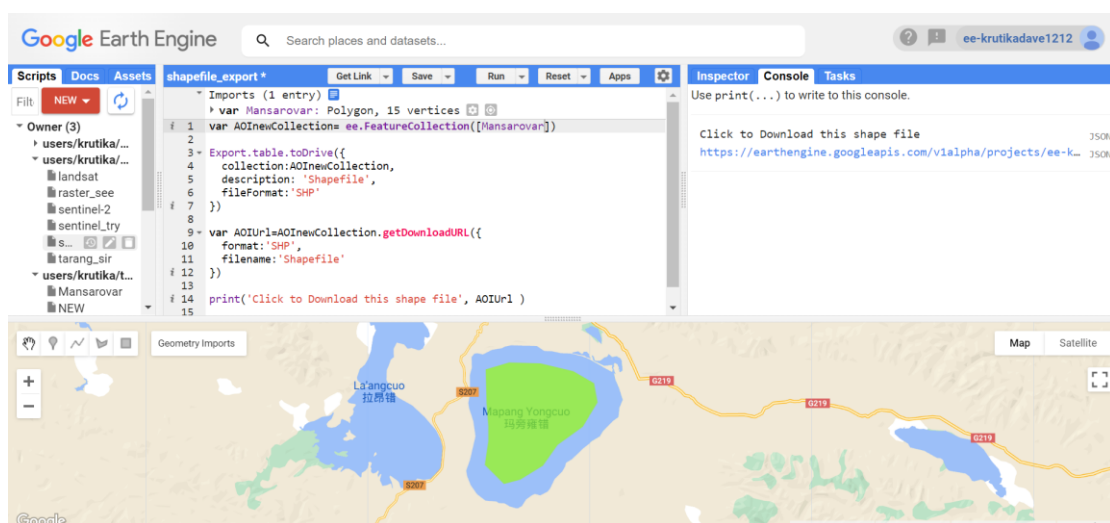


Figure 7: The script for exporting the polygon of the area of interest in the code editor section of Google Earth Engine.

Masking:

QGIS software was used for masking. The following are the steps involved:

1. Masking involves converting the vector file (AOI polygon) into a raster form.
 - Raster file represents spatial data in the form of a pixel wherein each pixel has a value stored in it that represents a specific physical property, such as Tb values in our SSMIS data.
 - Vector file represents spatial data as points, lines, and polygons, which are defined by their coordinates and associated attributes such as area, perimeter, etc.
2. A dialogue box appears. Inputs are added wherein the input layer is our AOI polygon and the reference layer is our SSMIS Tb data. Resolution of Tb data is entered in degrees i.e. 0.0625. The pixels in the dataset that lies within the polygon are supposed to be extracted and given a value of 1 (burn value) and the rest were marked 0 (no-data value). The values
3. The final raster file contains a few pixels that are confined within the AOI polygon.

Read the SSMIS data:

This was a two-step process performed in a Python environment.

1. Reading the SSMIS data file downloaded in .nc file format.
2. Using a for loop, the Tb values for the data that overlays with rasterized AOI are extracted, and the average value is then calculated and appended in an array. The final Tb data corresponding to all the years is generated that is further converted to an Excel file.

3. The 10-year data is divided into 4 sections (with a range of 3 years) in an Excel file for ease of understanding. A line chart is plotted and further analysed.

Determining LIP variables

The dates for the freeze-thaw events were extracted by analysing the Tb curve for each hydrological year from 2010 to 2022. Following are the LIP variables chosen depending on the study area and Tb trend observed:

- Freeze Onset: FO is the first day on which Tb increases sharply which will usually be in winter. Or it is detected by picking the date before the significant upturn of the Tb value is observed.
- Ice On: After FO, The Tb fluctuates for a few days till it reaches its maximum value for that year (peak) which is marked as Ice on. The time when the lake has reached its maximum ice cover.
- Freezing Duration: The number of days between freeze-onset and ice-on days is calculated.
- Melt Onset: After the peak, the first drop in the Tb value is considered as melt onset.
- Ice Off: The day of the year when the lake becomes totally ice-free is determined by the lowest value it reaches.
- Melting Duration: The number of days between Melt onset and Ice off is calculated.

Verification of methodology and results

- The same methodology was applied to Great Bear Lake and Qinghai Lake. A study published on these lakes was used as a reference and the Tb trend was compared to validate the procedure.
- Another research study reported average dates of all the LIP variables for the Mansarovar Lake and therefore the LIP determined in this study was verified with theirs[12].

Comparison of SSMIS LIP with other satellite-derived products

Due to the low temporal resolution of optical data, procuring satellite images for the exact dates of all the LIP variables becomes difficult. Therefore, two satellite data are used in support. Also, the high spatial resolution provides a detailed visual differentiation between ice and water. One hydrological year was selected (2018-2019) to show the optical imagery corresponding to the extracted freeze-thaw dates.

Google Earth Engine helps save time by providing top-of-atmosphere corrected optical images. The correction includes the pre-processing step that removes the effects of atmospheric scattering and absorption on the remotely sensed data. Thus, the optical images for this study were acquired from GEE.

Understand changes in Air temperature

The air temperature downloaded from the ERA5 platform was analysed in Excel through a line graph. The curve was combined with the Tb trend.

The results from SSMIS brightness temperature data, optical visual imagery and air temperature were combined and further interpretations were made.

Chapter V: Results and Discussion:

Analysis of one hydrological year (2018-2019)

Observations were noted from the trend of Tb and air temperature for one hydrological year (2018-2019) (Figure 8). A hydrological year is a standard period used in research studies. Here, August 2018 to July 2019 represents one water cycle.

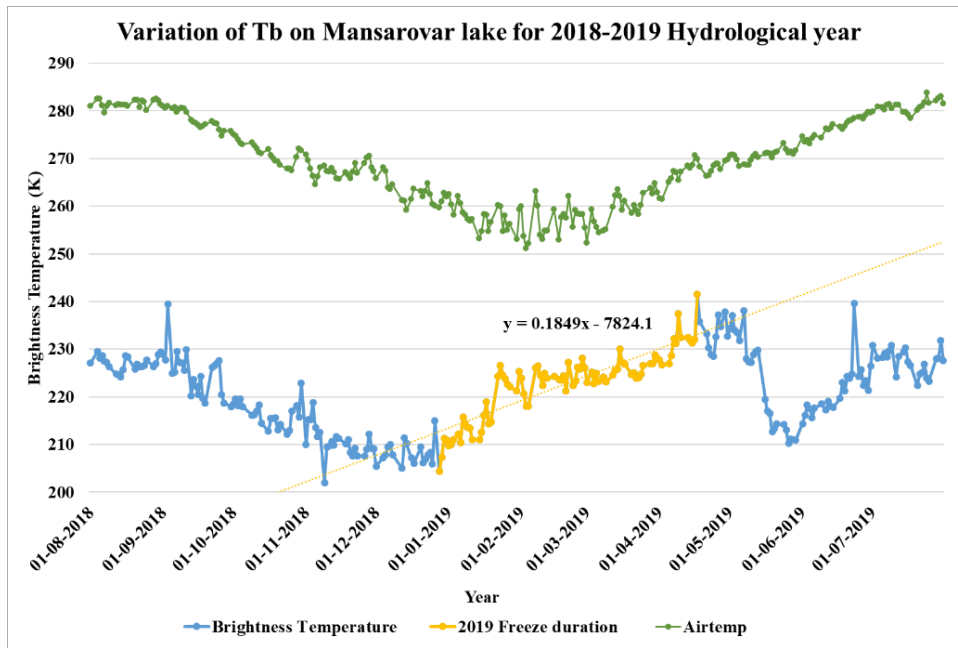


Figure 8: Variation of Tb for the 2018-2019 hydrological year highlighting the freezing duration

- In December, after the lowest drop in Tb value, the following rise is marked as Freeze onset i.e., 28th December 2018. The Tb then further increases and peaks on 17th April 2019 where the lake is expected to reach its maximum ice formation.
- The next drop in Tb implies the start of melting which is observed to be on 18th April 2019. The lake took 37 days to melt that year. The lowest value of

Tb is reached on 26th May 2019 thus the lake will be completely ice-free on that day.

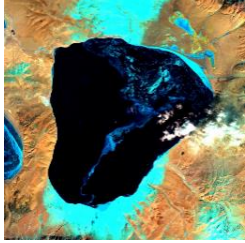
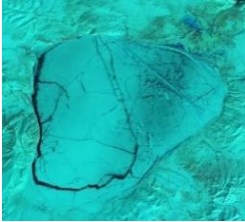
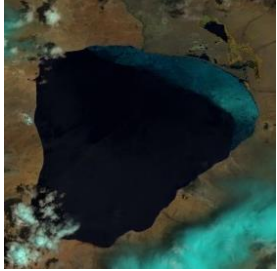
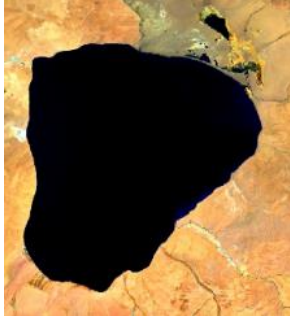
With reference to the brightness temperature equation ($TB = \varepsilon T_{kin}$), during the phase transition from ice to water or vice versa where the temperature remains stable, the emissivity factor plays a role. While once the phase is changed, the emissivity will remain constant and the change in air temperature will govern the Tb values. This relation is also reflected in the Figure 8.

- During the freezing period, the air temperature remains steady but the Tb is observed to be rising. This is due to the impact of higher emissivity of ice. Therefore, the increase in lake ice cover affects the Tb during this duration.
- Similarly, during the melting period, the impact of change in emissivity due to phase change is higher which is observed as a rapid decline in Tb value. Once the whole lake becomes ice-free, the emissivity factor in the equation becomes constant. And thereafter, the rise in Tb values is observed due to the rise in air temperature.

Comparison with Optical remote sensing data:

The extracted LIP for 2018-2019 is compared with the optical data procured for the nearest date possible. Optical image-based visual interpretation aligns with the LIP variables extracted from SSMIS Tb data for the 2018-2019 period (Figure 8 & Table 2).

Table 2: Comparison of LIP variables with the optical imagery for the 2018-2019 hydrological year

LIP variable	Satellite	Date	Optical Satellite Data
Freeze-onset	Sentinel-2	8 th Jan 2019	
Presence of snow during the freezing period	Landsat-8	24 th Jan 2019	
Ice-on	Landsat-8	14 th April 2019	
Melting of the lake during the Melting Duration	Landsat-8	16 th May 2019	
Ice-off	Sentinel-2	29 th May 2019	

LIP from 2010-2022:

The LIP variables are defined by analysing each hydrological year separately. The dates of each variable extracted are shown in Table 3.

- The Freezing duration is observed to be in the range of 89-144 days. The lake took maximum days to freeze (144) in the year 2013 and minimum (89 days) in the year 2021.
- Similarly, in the year 2011, the lake took 37 days to melt (highest) and 12 days in 2013 (lowest).

Table 3: LIP variables extracted from SSMIS Tb data from 2010-2022

Year	Freeze Onset (FO)	Ice-on (peak)	FD (days)	Slope	Melt Onset	Ice-off	MD (days)
2010-2011	25-Dec	12-Apr	108	0.22	23-May	23-May	37
2011-2012	29-Dec	02-Apr	95	0.22	02-May	02-May	26
2012-2013	22-Dec	15-May	144	0.18	17-May	29-May	12
2013-2014	30-Dec	04-May	96	0.19	07-May	02-Jun	26
2014-2015	30-Dec	13-May	134	0.13	18-May	04-Jun	17
2015-2016	25-Dec	30-Mar	96	0.24	31-Mar	27-Apr	27
2016-2017	29-Dec	01-Apr	93	0.20	02-Apr	03-May	30
2017-2018	07-Jan	11-Apr	94	0.26	12-Apr	01-May	19
2018-2019	28-Dec	17-Apr	110	0.19	18-Apr	26-May	37
2019-2020	26-Dec	20-Apr	116	0.17	22-Apr	09-May	17
2020-2021	30-Dec	29-Mar	89	0.18	31-Mar	16-Apr	16
2021-2022	28-Dec	07-Apr	100	0.19	12-Apr	03-May	21

Freeze Onset:

According to the Tb values, the initial ice formation starts at the end of December for all the years except in the year 2016-2017 which goes up to the 7th of Jan. The change in freeze onset over 10 years (Figure 9) shows an increasing trend line which indicates the delay in ice formation in recent years.

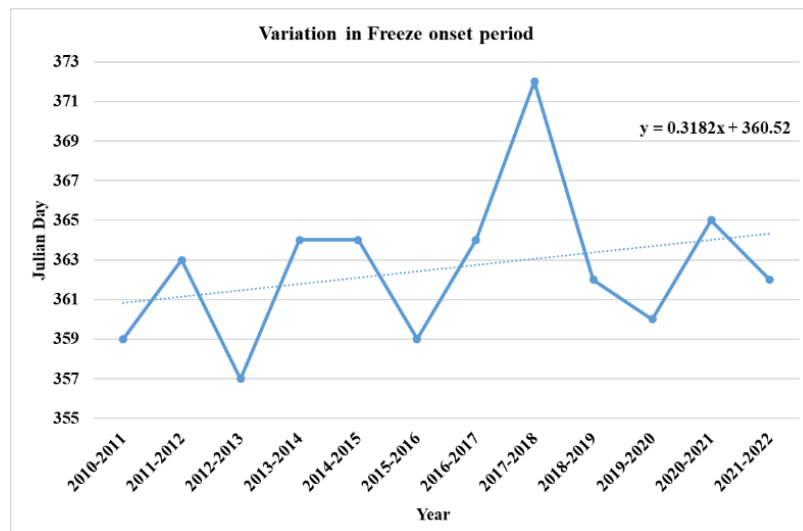


Figure 9: Variation in Freeze Onset from 2010-2022

Freezing Duration:

- The number of days between freeze onset and ice-on date signifies freezing period and its change shows a declining trend line indicating shortening of the freezing days (Figure 10).

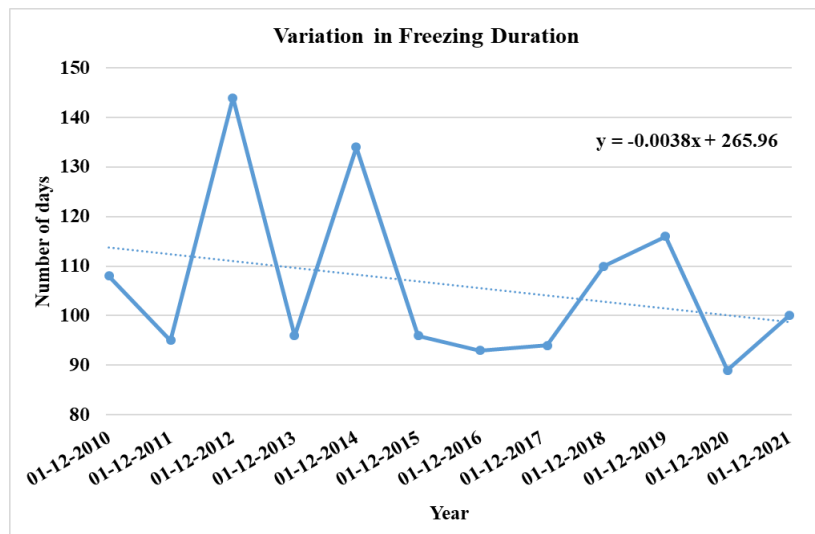


Figure 10: Variation in the Freezing Duration from 2010-2022

- Figure 11 shows the change in the rate of freezing over the years, calculated based on freezing duration. The decline in trend shows that the ice is forming at a slower rate compared to the early years (Figure 11).

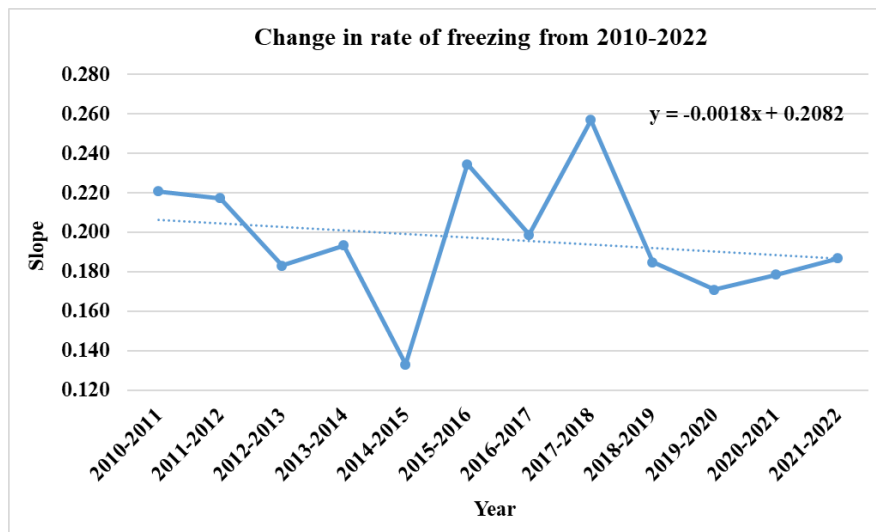


Figure 11: Variation in rate of freezing from 2010-2022

Melt Onset and Melting Duration:

A declining trend in melt onset over the years indicates the early melting of the lake in recent years (Figure 12). The decline in the melting period implies that the lake takes lesser days to melt completely (Figure 13).

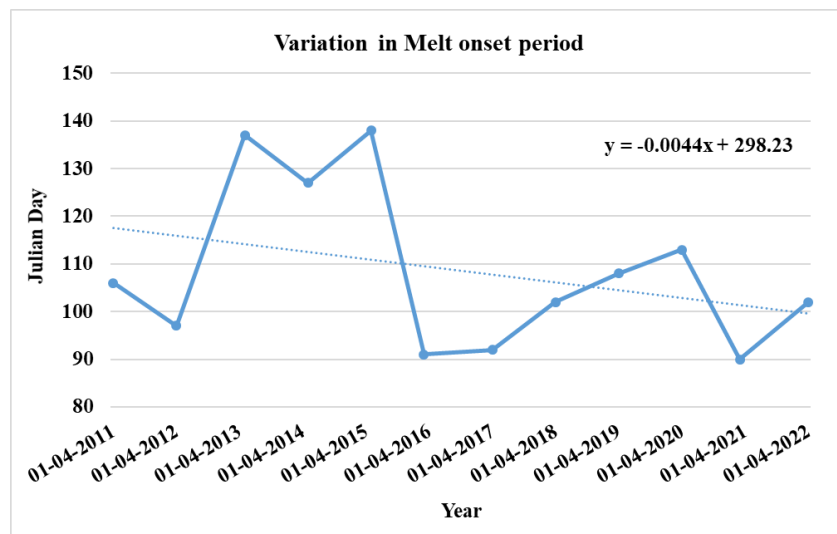


Figure 12: Variation in melt onset from 2010-2022.

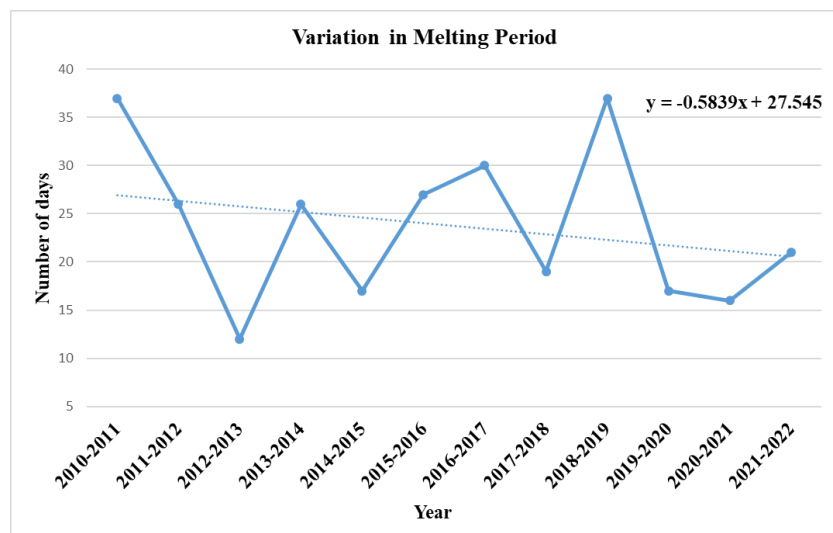


Figure 13: Variation in Melting Duration from 2010-2022.

The trend of Tb and air temperature 2010-2022

Figure 14,15,16,17 summarizes the variation in Brightness temperature and air temperature over 10 years. The freezing duration in each Tb curve is highlighted. Depending on the availability of air temperature data, the curve is plotted from the year 2011 to 2022.

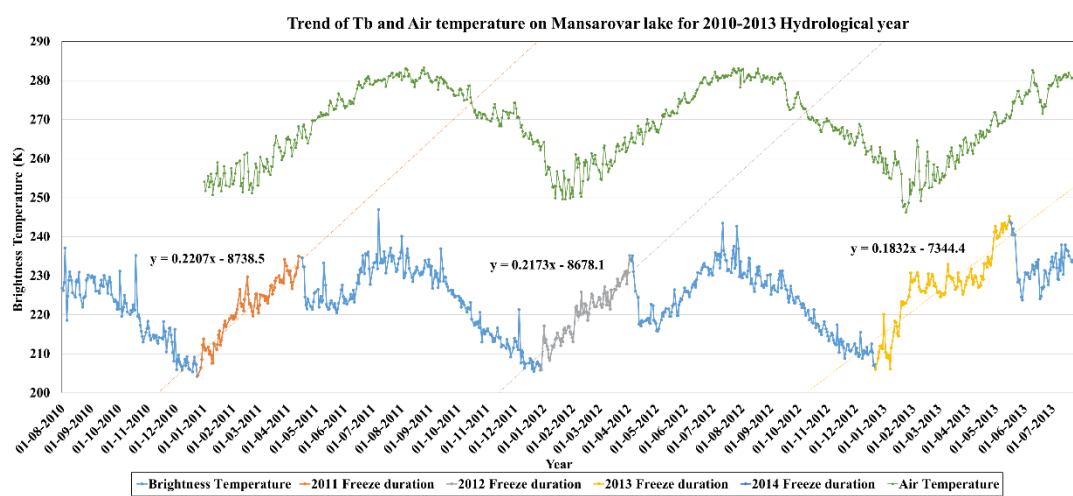


Figure 14: Variation of Tb and Air temperature observed from 2010-2013 and the Freezing Duration of each year is highlighted.

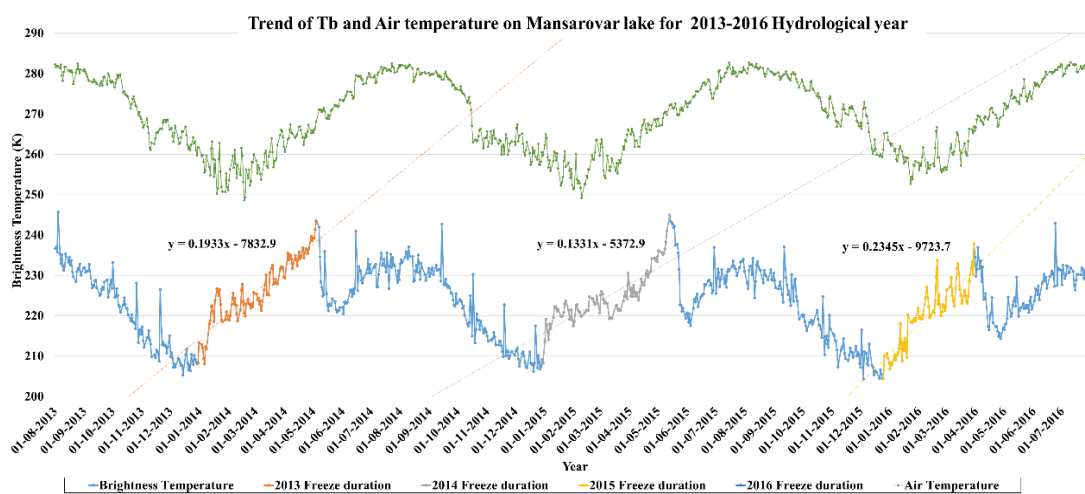


Figure 15: Variation of Tb and Air temperature observed from 2013-2016 and the Freezing Duration of each year is highlighted.

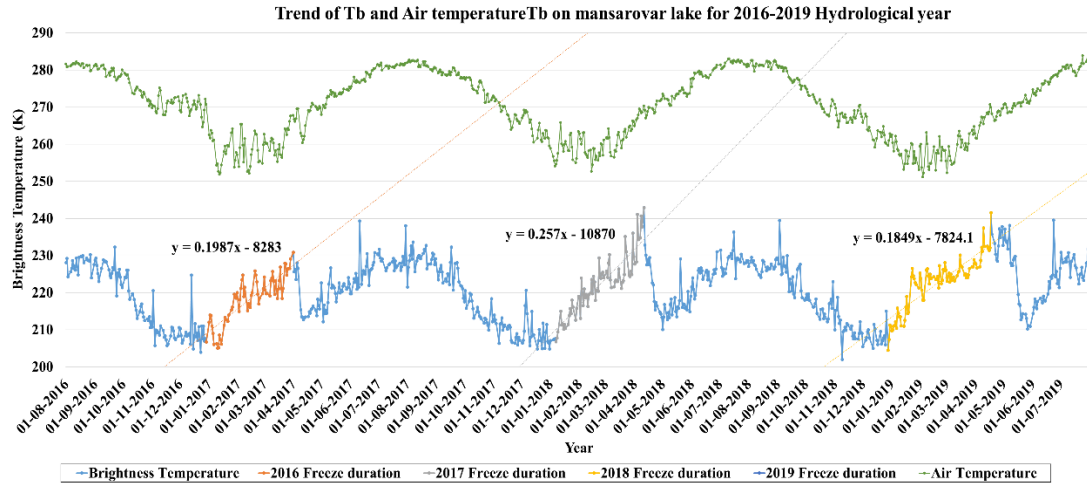


Figure 16: Variation of Tb and Air temperature observed from 2016-2019 and the Freezing Duration of each year is highlighted.

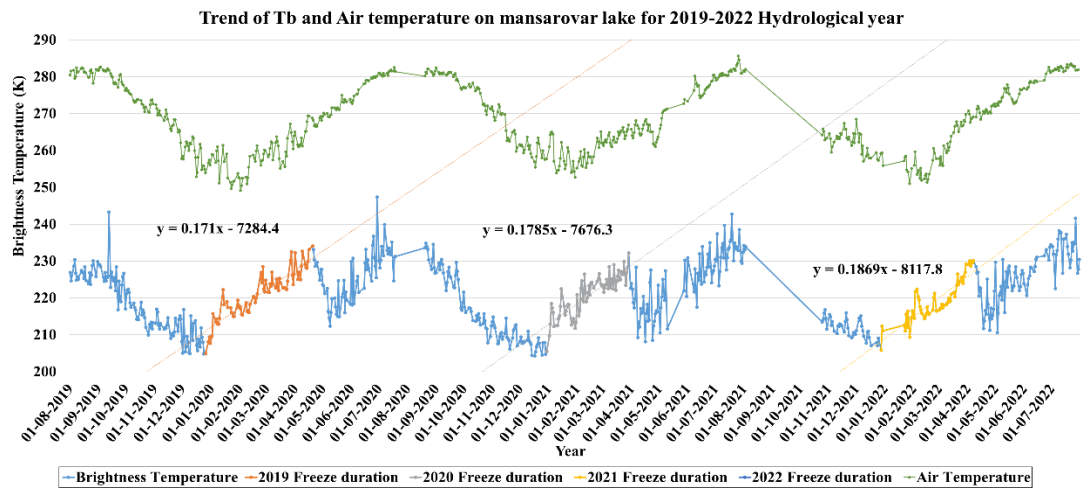


Figure 17: Variation of Tb and Air temperature observed from 2019-2022 and the Freezing Duration of each year is highlighted.

During the transition phase, when the lake is experiencing ice formation and melting, the brightness temperature shows a drastic change in its values while there is no

significant change in air temperature. Thus, rather than the values of air temperature, the rapid increase and decline in Tb values help determine the LIP variables.

Cross Validation:

1. The methodology of this study was validated with the results from two other studies performed on Lake Qinghai and Great Bear Lake. The Tb trend was compared and found to be showing a similar trend. The study period of the Qinghai Lake was possible to be kept similar for comparison i.e., 2014-2015 (Figure18,19)

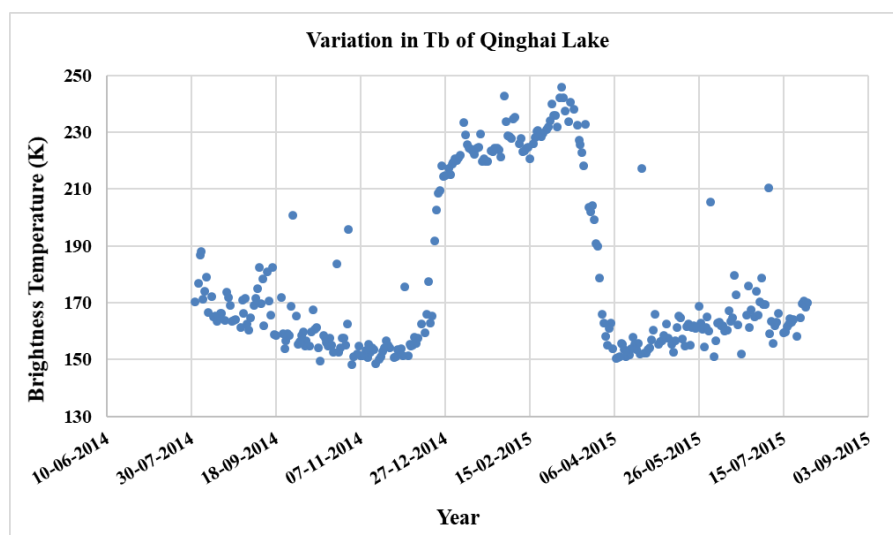


Figure 18: Change in Tb values for 2014-2015 Hydrological year for Qinghai Lake.

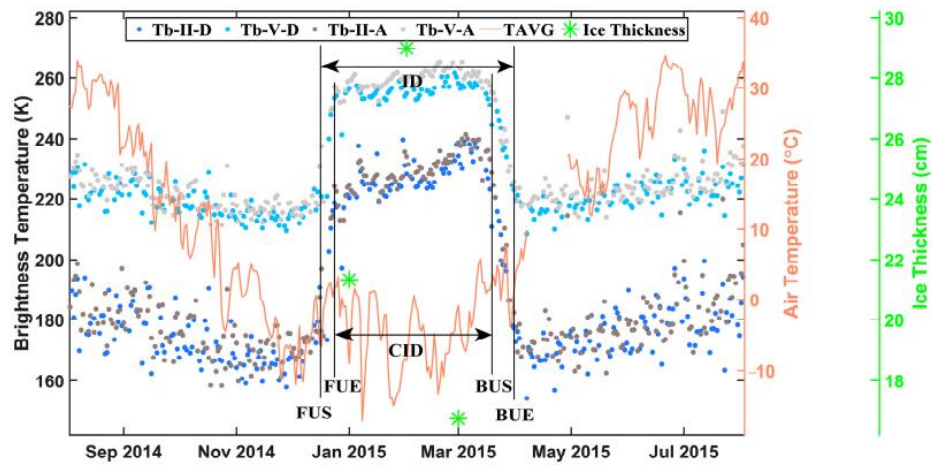


Figure 19: Change in Tb values for 2014-2015 Hydrological year for Qinghai Lake

(Source: Lei Su et al., 2021)

2. Another lake of interest was chosen, the Great Bear Lake and the Tb trend for 2010-2011 was observed. The pattern was found to be similar to that of the previous study done on the same lake for period 2003-2004 (Figure 20,21).

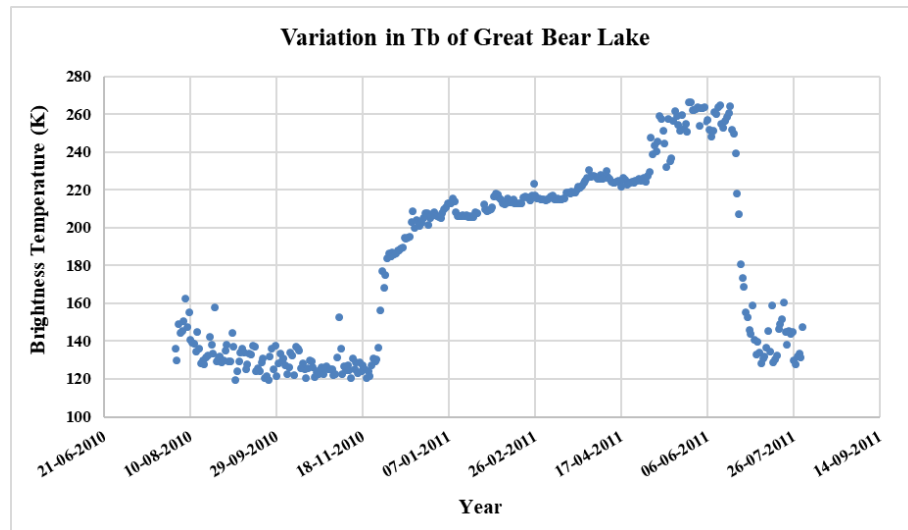


Figure 20: Change in Tb values for 2014-2015 Hydrological year for Great Bear Lake, Canada

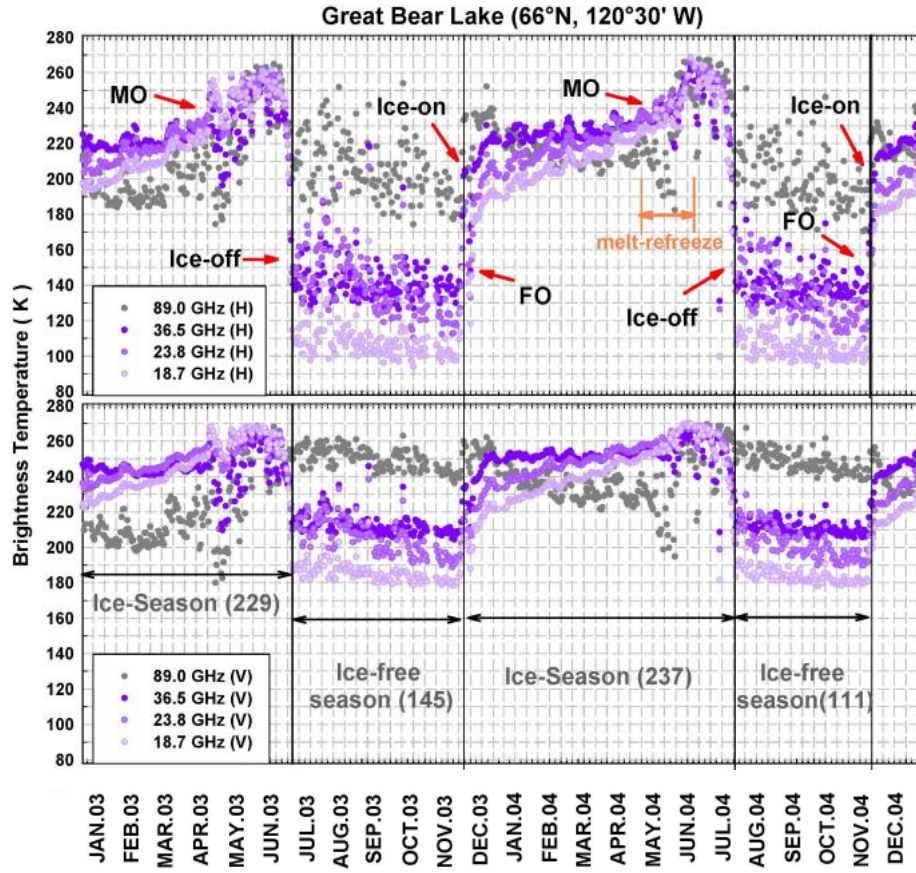


Figure 21: Change in Tb values for 2014-2015 Hydrological year for Great Bear Lake, Canada

(Source: K.-K Kang et al., 2012)

3. Besides, the FO, MO, and Ice-off dates in the current study are found to be at the end of Dec, Mid-April, and Mid May respectively which closely coincides with the results from a previous study done on 59 large lakes of the Tibetan Plateau using optical data[12]. They reported the mean FO of Mansarovar Lake to be on 9th Jan, MO on 15th April, and Ice-off on 5th May[12]. The results are not the same due to the different timeline (2001-2010) and dataset (optical data) used in the previous study.

Chapter VI: Conclusion

- The LIP of Mansarovar Lake was successfully determined using SSMIS Tb data for ~12 years. The comparison with the visual interpretation of optical data was found to be complimenting the LIP results obtained from SSMIS Tb data.
- Furthermore, positive observations were made during cross-validation of the methodology applied in this study with the previous studies using same AOI.
- The analysis of LIP variables shows delay in the freeze onset period, advance in melt onset, lesser freezing days and melting days. Such a shift could be because of multiple factors that influence the Tb value including air temperature, snowfall, and the effect of wind velocity.
- This study tries to fill the gap in the LIP database by choosing lakes that have less scientific data on their freeze-thaw events; such as the Mansarovar lake. Additionally, fewer studies have explored the latest enhanced 6.25km SSMIS grid product providing better spatial resolution. Thus this research project can potentially serve as a good primary base for working with the enhanced dataset on various other lakes.

Chapter VII: Future Aspects:

- Adding a larger temporal data to this study will provide a better understanding of the changes in Lake Ice Phenology. The change in the trend of Tb can be supported by analysing additional parameters like lake surface temperature, snowfall, and wind. Studying the impact of each of these factors on LIP variables shall provide us with a complete explanation of the changes observed over the years.
- The extraction of LIP variables was carried out manually which can be further automated through developing an algorithm.
- Once the study is performed on one lake, it could further be replicated over a larger area, especially the lakes of the Himalayan region like Tso Moriri, and Pangong Lake where scientific data are scarce.

Chapter VIII: References

1. Duguay, C., et al., *Ice-cover variability on shallow lakes at high latitudes: Model simulations and observations*. Hydrological Processes, 2003. **17**: p. 3465-3483.
2. Kang, K.-K., C.R. Duguay, and S.E.L. Howell, *Estimating ice phenology on large northern lakes from AMSR-E: algorithm development and application to Great Bear Lake and Great Slave Lake, Canada*. The Cryosphere, 2012. **6**: p. 235-254.
3. Wang, X., et al., *A lake ice phenology dataset for the Northern Hemisphere based on passive microwave remote sensing*. Big Earth Data, 2022. **6**(4): p. 401-419.
4. Su, L., T. Che, and L. Dai, *Variation in Ice Phenology of Large Lakes over the Northern Hemisphere Based on Passive Microwave Remote Sensing Data*. Remote Sensing, 2021. **13**(7): p. 1389.
5. Sharma, S., et al., *Loss of Ice Cover, Shifting Phenology, and More Extreme Events in Northern Hemisphere Lakes*. Journal of Geophysical Research: Biogeosciences, 2021. **126**(10): p. e2021JG006348.
6. Yongjian, R., et al., *Prediction and Analysis of Lake Ice Phenology Dynamics Under Future Climate Scenarios Across the Inner Tibetan Plateau*. Journal of Geophysical Research Atmospheres, 2020. **125**: p. 14.
7. Lang, J., et al., *The Impact of Climate Warming on Lake Surface Heat Exchange and Ice Phenology of Different Types of Lakes on the Tibetan Plateau*. Water, 2021. **13**: p. 634.

8. Yao, Z., et al., *Spatial-temporal patterns of major ion chemistry and its controlling factors in the Manasarovar Basin, Tibet*. Journal of Geographical Sciences, 2015. **25**.
9. Qi, M., et al., *Monitoring the Ice Phenology of Qinghai Lake from 1980 to 2018 Using Multisource Remote Sensing Data and Google Earth Engine*. Remote Sensing, 2020. **12**: p. 2217.
10. Justin Murfitt, C.R.D., *50 years of lake ice research from active microwave remote sensing: Progress and prospects*. Remote Sensing of Environment, 2021. **264**.
11. Yao, X., et al., *Spatial-temporal variations of lake ice phenology in the Hoh Xil region from 2000 to 2011*. Journal of Geographical Sciences, 2016. **26**: p. 70-82.
12. Kropáček, J., et al., *Analysis of ice phenology of lakes on the Tibetan Plateau from MODIS data*. The Cryosphere, Volume 7, Issue 1, 2013, pp.287-301, 2013. **7**: p. 287-301.
13. Sun, L., et al., *Analysis of Ice Phenology of Middle and Large Lakes on the Tibetan Plateau*. Sensors (Basel), 2023. **23**(3).
14. Woodhouse, I.H., *Introduction to microwave remote sensing*. 2006, The University of Edinburgh Scotland: CRC Press Taylor & Francis Group. 370.
15. Özkan, İ. and P. Baykal, *Investigation of Electromagnetic Polarization Properties of Silver-Containing Robotuft Carpets*. 2016.
16. Kok Yeow, Y., *Introductory Chapter: RF/Microwave Applications*, in *Emerging Microwave Technologies in Industrial, Agricultural, Medical and Food Processing*, Y. Kok Yeow, Editor. 2018, IntechOpen: Rijeka. p. Ch. 1.

17. Warren, S.G., *Optical properties of ice and snow*. Philos Trans A Math Phys Eng Sci, 2019. **377**(2146): p. 20180161.
18. Surdu, C.M., et al., *Ice Freeze-up and Break-up Detection of Shallow Lakes in Northern Alaska with Spaceborne SAR*. Remote Sensing, 2015. **7**(5): p. 6133-6159.
19. Murfitt, J. and C. Duguay, *Assessing the Performance of Methods for Monitoring Ice Phenology of the World's Largest High Arctic Lake Using High-Density Time Series Analysis of Sentinel-1 Data*. Remote Sensing, 2020. **12**: p. 382.
20. Yao, Z., et al., *Characteristics of isotope in precipitation, river water and lake water in the Manasarovar basin of Qinghai–Tibet Plateau*. Environmental Earth Sciences - ENVIRON EARTH SCI, 2009. **57**: p. 551-556.
21. Rai, S.P., et al., *Possibility of Hydrological Connectivity between Manasarovar Lake and Gangotri Glacier*. Current science, 2019. **116**: p. 1062-1067.
22. Brodzik, M.J., D. G. Long, M. A. Hardman, A. Paget, and R. Armstrong. (2016). *MEaSURES Calibrated Enhanced-Resolution Passive Microwave Daily EASE-Grid 2.0 Brightness Temperature ESDR, Version 1 [Data Set]*.